Long-term prediction of spatiotemporal chaotic systems using ensemble reservoir computing

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Introduction

Introduction

Spatiotemporal chaos and the prediction challenge
Spatiotemporal chaotic systems, governed by nonlinear PDEs, exhibit exponential divergence of nearby trajectories - placing a hard theoretical limit on deterministic predictability. Long-term prediction necessarily operates in a **closed-loop (autonomous) regime**, making error accumulation inevitable and rapid collapse the norm for most ML approaches.

Reservoir Computing as a structured alternative
Reservoir Computing (RC) offers a theoretically grounded alternative: only the linear readout is trained via ridge regression, while the fixed random reservoir provides the echo state property and noise robustness. Yet RC-based long-term prediction of **high-dimensional** spatiotemporal chaotic systems remains insufficiently studied.

Our goal & approach
We target two failure modes that emerge when scaling RC to high dimensions over long horizons: **(1) noise-driven collapse** and **(2) autonomous prediction instability** - periodicity locking, phase inversions, and decoherence. We propose two complementary ensemble RC strategies, demonstrated on the KS equation (128D), aimed at maintaining **long-term statistical validity** well beyond the deterministic predictability horizon.

Problem Setting

The Kuramoto–Sivashinsky (KS) Equation

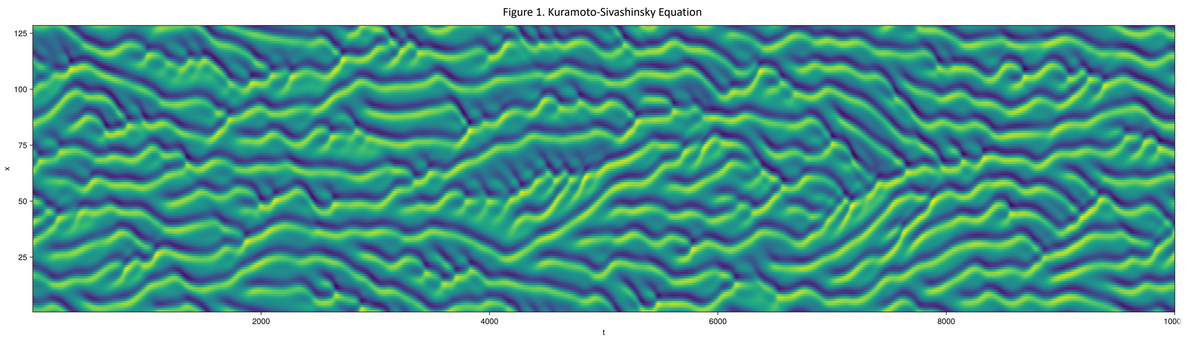
$$\partial_t u + u\partial_x u + \partial_x^2 u + \partial_x^4 u = 0, x \in [0, L]$$

With Additive Gaussian and Measerment noise:

$$\tilde{u}(x, t) = u(x, t) + \varepsilon, \varepsilon \sim \mathcal{N}(0, \sigma^2 I)$$

Parameter	Value
Domain size	$L = 2\pi K, K = 15$
Spatial points	128
Fourier modes	$N = 40$
Time step	$h = 1/15$
Total steps	$T = 100{,}000$

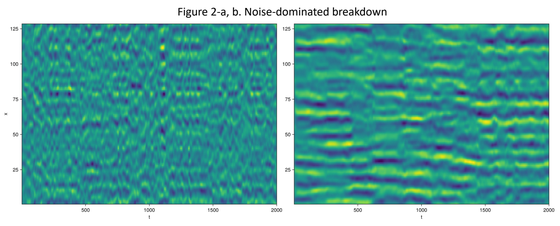
Figure 1. Kuramoto-Sivashinsky Equation



Challenges in long-term prediction

Failure Mode 1: Noise-driven Collapse

Figure 2-a, b. Noise-dominated breakdown



Failure Mode 2: Single-reservoir Instability over Long Horizons

Figure 3-a. Flip

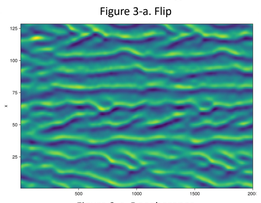


Figure 3-b. Damping

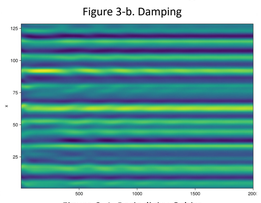


Figure 3-c. Decoherence

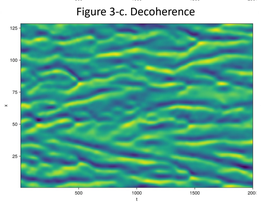
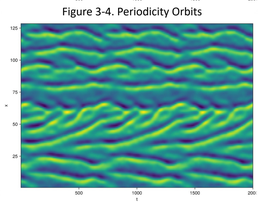


Figure 3-4. Periodicity Orbits



Methods

Reservoir Computing

Architecture
An Echo State Network (ESN) consists of three components: an input layer, a fixed randomly initialized sparse recurrent reservoir, and a trained linear readout layer. Formally, the reservoir state evolves as:

$$\mathbf{r}(t+1) = \tanh\left(W_{res}\,\mathbf{r}(t) + W_{in}\,\mathbf{u}(t)\right)$$

where W_{res} is the fixed reservoir weight matrix and W_{in} is the fixed input weight matrix. Only the readout W_{out} is trained via ridge regression:

$$W_{out} = \arg\min_{W} \sum_t \| W\mathbf{r}(t) - \mathbf{u}(t+1) \|^2 + \beta \| W \|^2$$

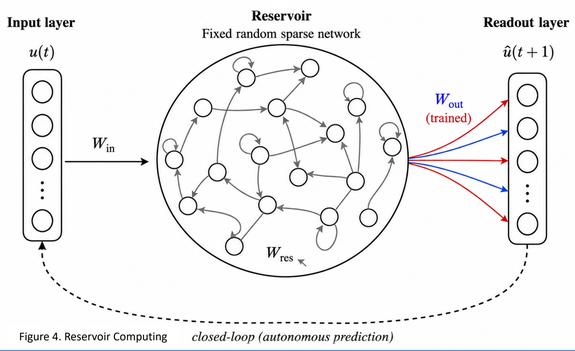


Figure 4. Reservoir Computing

Proposed Methods

Method 1:
Step-by-step Mean Ensemble (SME) Reservoir

Method 2:
Iterative Rollout Ensemble (IRE) Reservoir

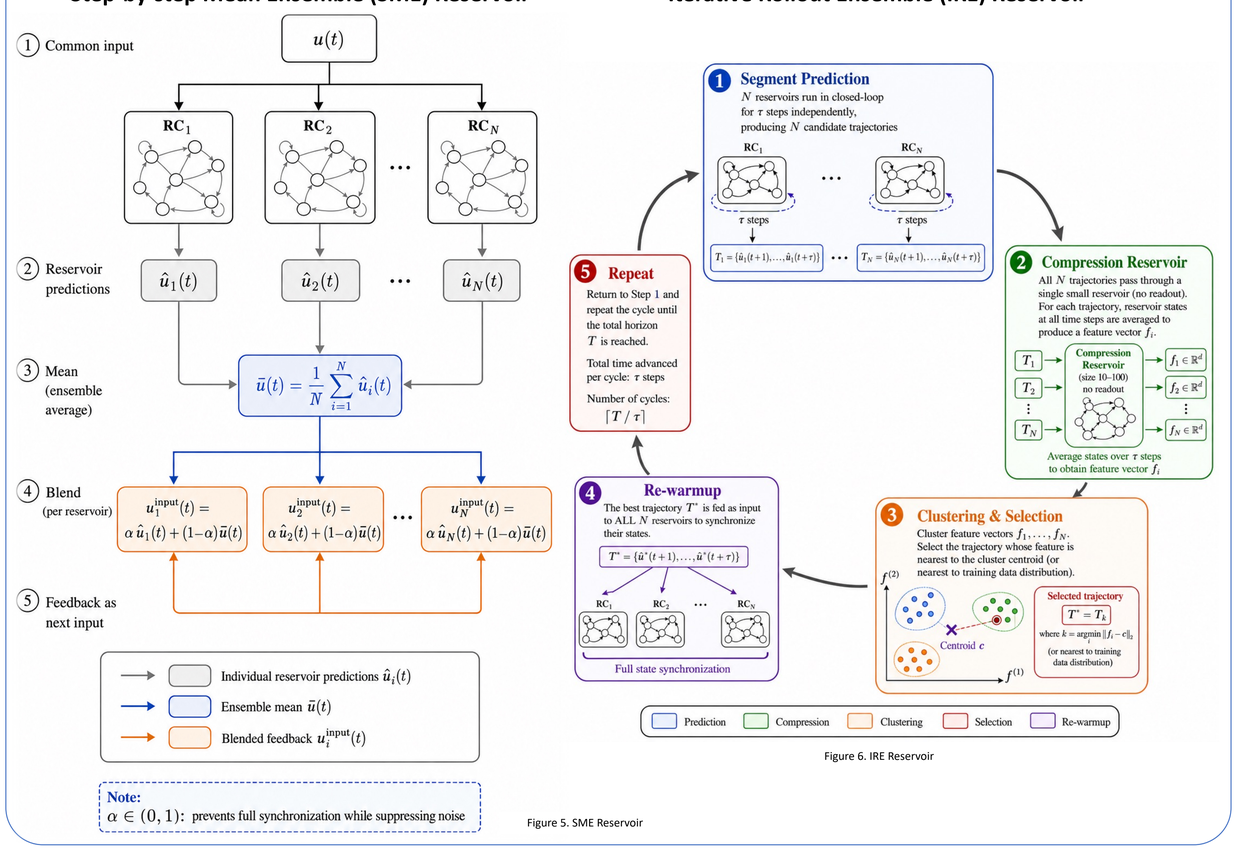


Figure 5. SME Reservoir

Figure 6. IRE Reservoir

Results

Long-term prediction quality - Method 1 (SME)

Figure 6-a. Ground Truth

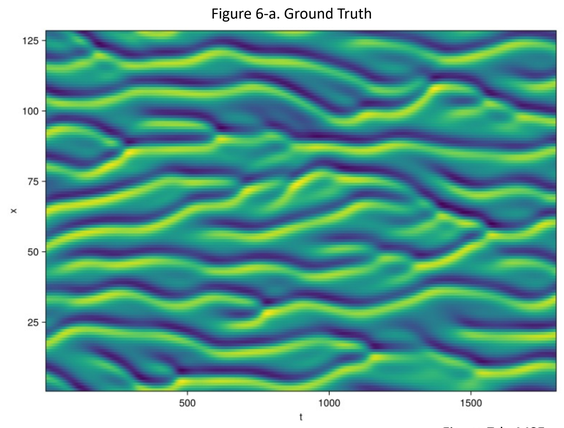


Figure 6-b. Single Reservoir

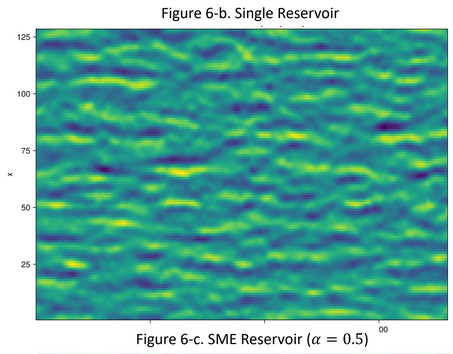


Figure 6-c. SME Reservoir ($\alpha = 0.5$)

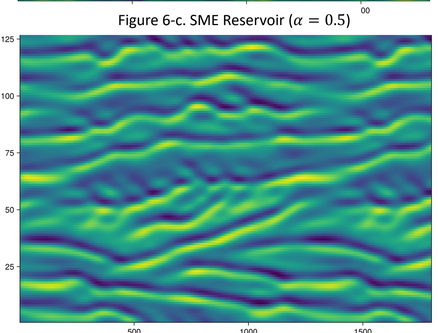


Figure 7-a. VPT

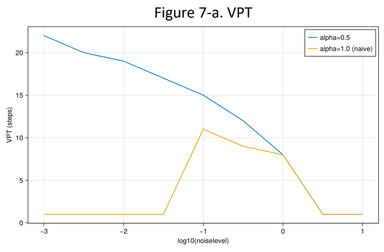


Figure 7-b. MSE

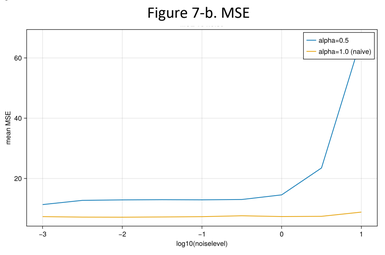
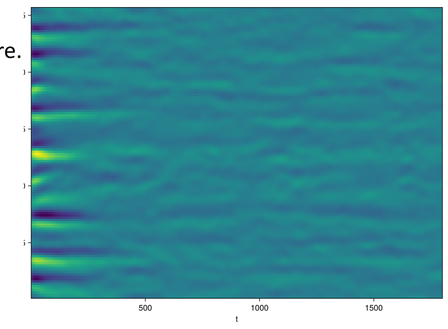


Figure 6-d. Naïve Ensemble Reservoir ($\alpha = 1.0$)



Under high observation noise, single RC predictions rapidly degenerate into incoherent states, failing to capture any meaningful spatiotemporal structure. Qualitatively, SME produces trajectories most visually consistent with the ground truth among all compared methods. While the Naive Ensemble achieves lower MSE than SME, this is misleading — the Naive Ensemble reduces error by averaging toward a mean state, suppressing dynamic variability rather than capturing the true chaotic trajectory. VPT analysis confirms this: the Naive Ensemble fails to reproduce the underlying dynamics despite its lower pointwise error. SME, by contrast, preserves dynamical structure through diversity-maintaining blending, achieving superior long-term prediction quality.

Long-term statistical validity - Method 2 (IRE)

Figure 8-a. Ground Truth

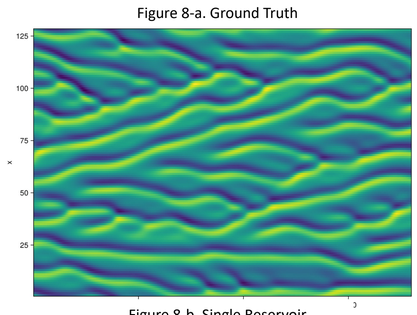


Figure 8-b. Single Reservoir

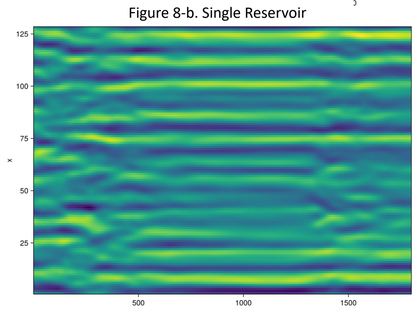


Figure 8-c. IRE Reservoir

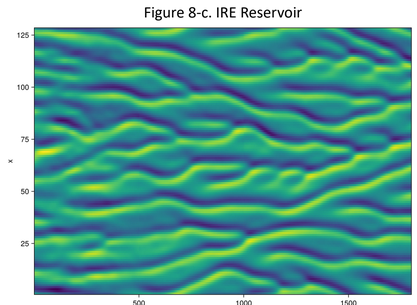


Figure 8-d. SME Reservoir

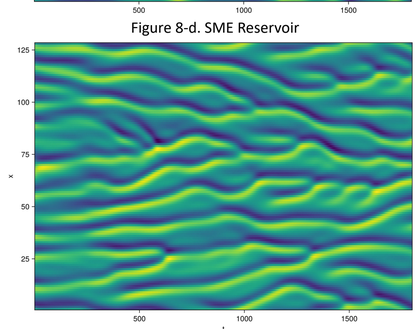


Figure 9-a. Mean

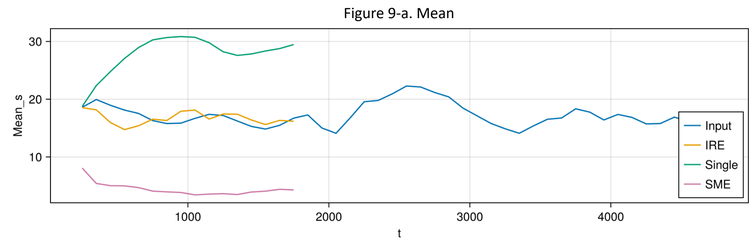


Figure 9-b. Standard Deviation

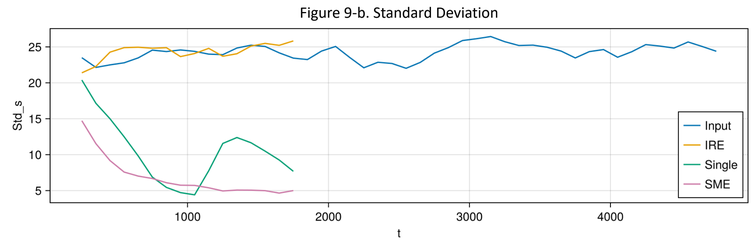


Figure 9-c. Covariance

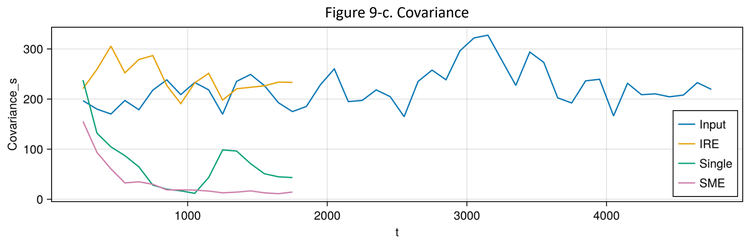
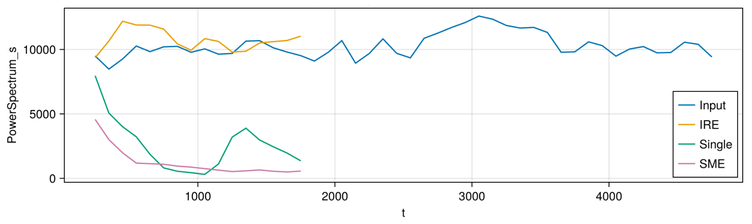


Figure 9-d. Power Spectrum



In the low-noise regime, the primary challenge is single-reservoir instability over long horizons. Both SME and IRE qualitatively outperform single RC, producing trajectories that remain coherent over extended prediction windows. To quantify long-term statistical validity, we apply a sliding time window to each predicted trajectory and compute statistics including mean, standard deviation, and covariance, comparing them against those of the ground truth. IRE consistently maintains statistical properties closest to the input trajectory throughout the prediction horizon, demonstrating its effectiveness in sustaining physically meaningful long-term dynamics.

Conclusion

We proposed two complementary ensemble RC strategies for long-term prediction of high-dimensional spatiotemporal chaos

Method 1 (Step-by-step mean ensemble):

- Suppresses noise-driven collapse via mean-blending with partial feedback, preserving ensemble diversity

Method 2 (Iterative rollout ensemble):

- Maintains long-term prediction quality by iteratively selecting physically consistent trajectories using compression reservoir-based clustering
- The compression reservoir provides an efficient, training-free feature extractor for trajectory quality assessment

Future work

- adaptive blending parameter α,
- learned trajectory selection,
- extension to other chaotic PDEs (e.g., 2D Navier–Stokes)